19.19 Gradient

As though there were not enough notation associated with vector functions, there is a further notation which is used for the derivative of a real-valued function. A derivative $f'(\xi)$ in this case is a $1 \times n$ matrix, i.e. a row vector. When matrix notation is not being systematically used especially in applied mathematics, one sometimes writes

$$\nabla f(\boldsymbol{\xi}) = f'(\xi)$$

and refers to $\nabla f(\boldsymbol{\xi})$ as the *gradient* of f at $\boldsymbol{\xi}$.

The geometric interpretation of the gradient (i.e. the derivative) of a real-valued function of a vector variable is worth noting. Recall that

$$y - f(\xi) = f'(\xi)(x - \xi)$$

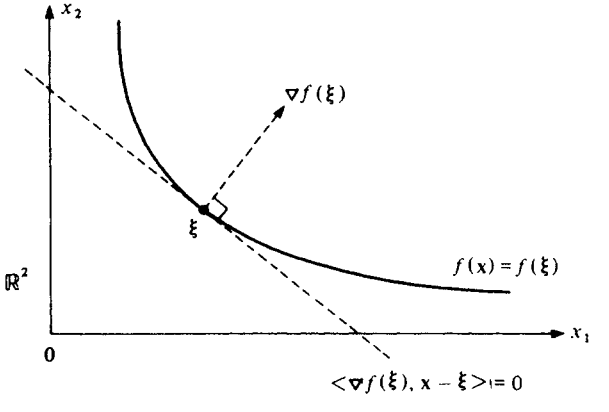
is the equation of the hyperplane in $\mathbb{R}^{n+1}$ which is the tangent to $y = f(x)$ where $x = \xi$. It follows that

$$f'(\xi)(x - \xi) = 0$$

is the equation of the hyperplane in $\mathbb{R}^n$ which is the tangent to the contour $f(x) = f(\xi)$ where $x = \xi$. Rewriting the equation $f'(\xi)(x - \xi) = 0$ in the form

$$\langle \nabla f(\boldsymbol{\xi}), \mathbf{x} - \boldsymbol{\xi} \rangle = 0$$

and referring to §18.15, we observe that $\nabla f(\boldsymbol{\xi})$ is a *normal* to the contour $f(\mathbf{x}) = f(\boldsymbol{\xi})$ at $\mathbf{x} = \boldsymbol{\xi}$.



This provides a geometric interpretation for the direction in which $\nabla f(\boldsymbol{\xi})$ points. Its length can also be interpreted geometrically. The slope (or rate of increase) of f at $\boldsymbol{\xi}$ in the direction of the unit vector $\mathbf{u}$ is given by

$$D_{\mathbf{u}} f(\boldsymbol{\xi}) = f'(\xi)u = \langle \nabla f(\boldsymbol{\xi}), \mathbf{u} \rangle.$$

By the Cauchy–Schwarz inequality,

$$|D_{\mathbf{u}}f(\boldsymbol{\xi})| = |\langle \nabla f(\boldsymbol{\xi}), \mathbf{u}\rangle| \leqslant \|\nabla f(\boldsymbol{\xi})\|.\|\mathbf{u}\| = \|\nabla f(\boldsymbol{\xi})\|.$$

But, taking $\mathbf{u} = \|\nabla f(\boldsymbol{\xi})\|^{-1}\nabla f(\boldsymbol{\xi})$, we obtain that

$$D_{\mathbf{u}}f(\boldsymbol{\xi}) = \|\nabla f(\boldsymbol{\xi})\|.$$

Thus $\nabla f(\boldsymbol{\xi})$ points in the direction in which the slope of f at $\boldsymbol{\xi}$ is *greatest* and the magnitude of the slope in this direction is equal to $\|\nabla f(\boldsymbol{\xi})\|$.

19.20 Example Consider the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ of example 19.18(i). We have that

$$\nabla f(2, 1) = f'(2, 1) = (5, 10).$$

It follows that (5, 10) is a normal to the contour $x_1^2x_2 + x_2^3x_1 = 6$ where $(x_1, x_2) = (2, 1)$. Also (5, 10) is the direction in which f increases fastest at (2,1) and the rate of increase in this direction is $\{5^2 + 10^2\}^{1/2} = 5\sqrt{5}$.

19.21 Manipulation of derivatives

For real-valued functions of a real variable, the following formulae are familiar.

$$\text{(I)} \quad \frac{d}{dx}(Lx + c) = L$$

$$\text{(II)} \quad \frac{d}{dx}(Lx^2) = 2Lx$$

$$\text{(III)} \quad \frac{d}{dx}(Ly + Mz) = L\frac{dy}{dx} + M\frac{dz}{dx}$$

$$\text{(IV)} \quad \frac{d}{dx}(yz) = z\frac{dy}{dx} + y\frac{dz}{dx}$$

$$\text{(V)} \quad \frac{dz}{dx} = \frac{dz}{dy}\cdot\frac{dy}{dx}$$

$$\text{(VI)} \quad \frac{dy}{dx} = \left(\frac{dx}{dy}\right)^{-1}.$$

The range of validity of these formulae was discussed in chapter 10. It is a very convenient fact that versions of all these formulae remain valid when the small letters are re-interpreted as column vectors and the large letters as matrices, *provided* that the formula obtained after this re-interpretation makes sense. With this re-interpretation, for example, formula (V) asserts that one matrix is the

product of two other matrices and formula (VI) asserts that one matrix is the inverse of another. (The latter assertion, of course, makes no sense unless

$$\frac{dy}{dx}$$

is a square, non-singular matrix which requires in particular that x and y have the same number of co-ordinates.)

We leave further consideration of (V) and (VI) until the next section and focus attention for the moment on the other formulae. As far as these are concerned, (I) and (III) are straightforward to interpret and easy to prove (exercise 19.26(4)). Formulae (II) and (IV), however, make no immediate sense when x, y and z are interpreted as column vectors. We therefore re-arrange the formulae slightly so that they do make sense and obtain

$$\text{(II)}' \quad \frac{d}{dx}(x^{\mathrm{T}}Ax) = x^{\mathrm{T}}(A^{\mathrm{T}} + A)$$

$$\text{(IV)}' \quad \frac{d}{dx}(y^{\mathrm{T}}z) = z^{\mathrm{T}}\frac{dy}{dx} + y^{\mathrm{T}}\frac{dz}{dx}.$$

As an example, we shall deduce (II)′ from (IV)′. The proof of the latter formula is left as an exercise (19.26(4)).

19.22 Example Observe to begin with that (II)′ makes no sense unless A is a *square* matrix. From formula (IV)′ and (I),

$$\frac{d}{dx}x^{\mathrm{T}}Ax = \frac{d}{dx}x^{\mathrm{T}}(Ax) = (Ax)^{\mathrm{T}}\frac{dx}{dx} + x^{\mathrm{T}}\frac{d}{dx}(Ax)$$

$$= x^{\mathrm{T}}A^{\mathrm{T}}I + x^{\mathrm{T}}A = x^{\mathrm{T}}(A^{\mathrm{T}} + A).$$

Here I denotes the *identity matrix*, i.e.

$$I = \begin{pmatrix} 1 & 0 \ldots 0 \\ 0 & 1 \ldots 0 \\ \vdots & \\ 0 & 0 \ldots 1 \end{pmatrix}.$$

Note that in the special case when A is *symmetric*, i.e. $A^{\mathrm{T}} = A$, we have that

$$\frac{d}{dx}(x^{\mathrm{T}}Ax) = 2Ax.$$
